

Unconfoundedness II: Propensity Scores and Regression

BUSS975: Causal Inference in Financial Research

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Outline

1. Propensity scores
2. Regression
3. What matching cannot do
4. The NSW demonstration

1. Propensity Scores

Defining the propensity score

- **Propensity score:** $e(X) = \Pr(D = 1 \mid X)$
 - ▶ “the conditional probability of assignment to a particular treatment given a vector of observed covariates”
- **Balancing property** (Rosenbaum–Rubin 1983): $X \perp D \mid \Pr(D = 1 \mid X)$
 - ▶ $E[X \mid D = 1, \Pr(D = 1 \mid X)] = E[X \mid D = 0, \Pr(D = 1 \mid X)]$
 - ▶ Covariate distributions are similar between treatment and control conditional the propensity score.
- **Propensity score theorem** (Rosenbaum–Rubin 1983): if $(Y^1, Y^0) \perp\!\!\!\perp D \mid X$, then

$$(Y^1, Y^0) \perp\!\!\!\perp D \mid e(X)$$

- ▶ Conditioning on a **single number** does what conditioning on all of X does. Ten covariates collapse to one scalar — the curse of dimensionality defeated, for *conditioning*.

Estimating the score

- You know the **true** score only when you ran the experiment yourself. Otherwise, it must be estimated: $\Pr(D_i=1 \mid X_i) = F(\beta X_i)$, fitted by **logit or probit**.
- Judge the specification by **balance achieved**, not by fit. A score with a high pseudo- R^2 but poor balance is a bad score; the reverse is fine.
- Fit, check balance, add interactions and polynomials where balance fails, refit. A legitimate specification search, because it never looks at Y — the whole stage is **blind to the outcome**, which protects against fishing. Say so in the paper.
- **The score creates no support that X lacks.** If a covariate cell is all-treated or all-control, collapsing X to a scalar hides that; it does not fix it.

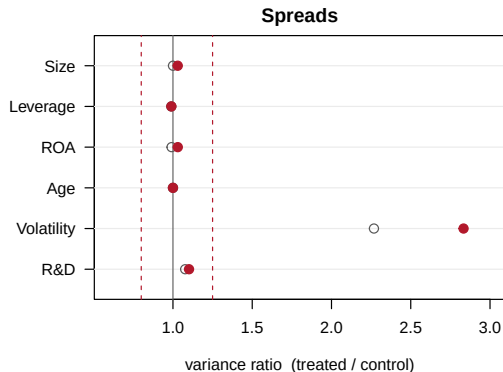
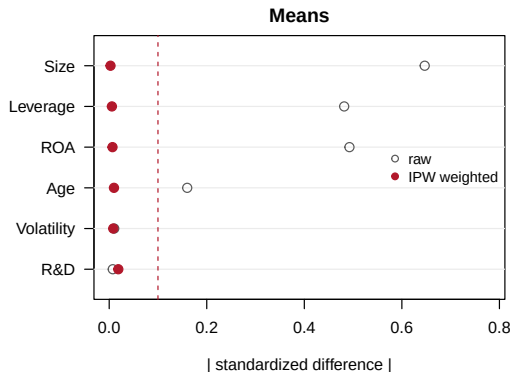
Checking balance: two numbers, not a t -test

The balancing property is checkable. Report it covariate by covariate, before and after.

$$\text{SMD} = \frac{\bar{X}_1 - \bar{X}_0}{\sqrt{(s_1^2 + s_0^2)/2}}, \quad \text{VR} = \frac{s_1^2}{s_0^2}$$

- **Standardized mean difference.** $|\text{SMD}| < 0.1$ is good.
 - ▶ Keep the denominator **fixed at its raw-sample value** when recomputing after weighting, or the yardstick moves with the thing you are measuring.
- **Variance ratio.** Means can match while spreads do not. Rubin: stay inside $[0.5, 2]$, ideally $[0.8, 1.25]$.
- **Do not use t -tests.** Balance is a property of *this sample*, not a population hypothesis — and p confounds imbalance with sample size (Imai, King and Stuart 2008).

A balance plot: means are not enough



Selection here is linear in Size, Leverage, ROA, Age — and **quadratic in Volatility**. The score was fitted with linear terms only.

Reading the balance plot

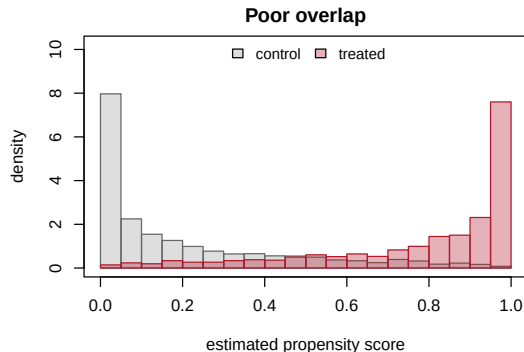
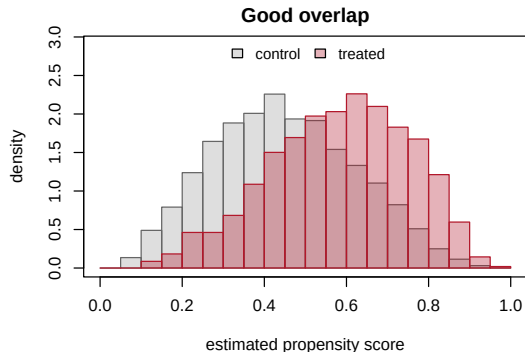
- **Left panel, the success.** Four covariates start badly imbalanced — Size at 0.65, Leverage and ROA near 0.49. After weighting, every one is under 0.02. This is the plot referees want, and it is the one most papers stop at.
- **Left panel, the trap.** Volatility looks *perfect* from the start: $SMD = -0.010$, and a t -test returns $p = 0.72$. On the evidence of means, nothing to see.
- **Right panel.** Its variance ratio is **2.27** — treated firms have more than twice the spread. Extreme volatility drives selection in *both* tails.
- **Weighting makes it worse**, $2.27 \rightarrow 2.83$, because the fitted score never contained vol^2 . Add that one term and the ratio falls to **1.03**.
- The lesson of this section: judge the score by **balance achieved, not by fit** — and balance means the whole distribution. Check squares and interactions, not just levels.

Three things to do with the score

Once every unit has a $\hat{e}(X_i)$, there are three standard uses.

1. **Diagnose.** Plot $\hat{e}$ separately for treated and control units. This is the common-support check, and it costs nothing.
2. **Weight.** Reweight the sample to build a pseudo-experiment — inverse probability weighting.
3. **Match.** Pair treated to control units with similar scores, then difference — propensity score matching.

Visualizing overlap



Right panel: a treated mass at $\hat{e} \approx 1$ with no controls, and a control mass at $\hat{e} \approx 0$ with no treated. Those units have **no counterfactual**. Any estimate covering them is extrapolation, not comparison.

This plot belongs in every such paper

- Common support is verifiable, unlike unconfoundedness.
- What to do when it looks like the right panel:
 - ▶ **Trim** observations with extreme scores, or impose a caliper. Drop units with $\hat{e} \notin [0.1, 0.9]$. A rule of thumb, but a defensible and widely accepted one (Crump, Hotz, Imbens and Mitnik, 2009).
 - ▶ **Report the change in estimand.** After trimming you describe a subpopulation, not the original one. Say how many observations went and how the treated sample shifted.
 - ▶ Consider retreating to the *ATT*, which only needs overlap in one direction.
- What *not* to do: report the estimate and stay silent. The regression will happily extrapolate into a region with no data and give you a standard error that reflects none of that.

Inverse probability weighting

Create balanced pseudo-population by inverse probability weighting. Write $\hat{e}_i$ for $\hat{e}(X_i)$ and $\pi = \Pr(D=1)$.

$$ATE = E[Y_1 - Y_0] = E\left[\frac{D Y}{e(X)}\right] - E\left[\frac{(1-D) Y}{1-e(X)}\right]$$

$$ATT = E[Y_1 - Y_0 | D = 1] = \frac{1}{\pi} \left(E[DY] - E\left[\frac{e(X)(1-D)Y}{1-e(X)}\right] \right)$$

$$\widehat{ATE}_{IPW} = \frac{1}{n} \sum_{i=1}^n \left[\frac{D_i}{\hat{e}_i} - \frac{1-D_i}{1-\hat{e}_i} \right] Y_i$$

$$\widehat{ATT}_{IPW} = \frac{1}{n_T} \sum_{i=1}^n \left[D_i - (1-D_i) \frac{\hat{e}_i}{1-\hat{e}_i} \right] Y_i$$

- Both are **Horvitz–Thompson (1952)** forms.

Why *that* weight for the ATE?

We prove $E\left[\frac{D \cdot Y}{e(X)}\right] = E[Y^1]$. The $E[Y^0]$ half is symmetric.

$$E\left[\frac{D \cdot Y}{e(X)}\right] = E\left[\frac{D \cdot Y^1}{e(X)}\right]$$

switching equation (consistency): $D \cdot Y = D \cdot Y^1$

$$= E_X\left[E\left(\frac{D \cdot Y^1}{e(X)} \mid X\right)\right]$$

law of total expectation

$$= E_X\left[\frac{1}{e(X)} \cdot E(D \cdot Y^1 \mid X)\right]$$

$e(X)$ is fixed given X

$$= E_X\left[\frac{1}{e(X)} \cdot E(D \mid X) \cdot E(Y^1 \mid X)\right]$$

unconfoundedness (exchangeability): $D \perp\!\!\!\perp Y^1 \mid X$

$$= E_X\left[\frac{1}{e(X)} \cdot e(X) \cdot E(Y^1 \mid X)\right]$$

by definition, $E(D \mid X) = e(X)$

$$= E_X[E(Y^1 \mid X)] = E[Y^1]$$

cancel; law of total expectation

Why *that* weight for the ATT?

$$E[Y^1 \mid D=1] = E[Y \mid D=1] = \frac{E[DY]}{E[D]}.$$

$$E[Y^0 \mid D=1] = E_X \left[E(Y^0 \mid X, D=1) \mid D=1 \right]$$

law of total expectation, over $f(x \mid D=1)$

$$= E_X \left[E(Y^0 \mid X, D=0) \mid D=1 \right]$$

unconfoundedness — swap the arm

$$= E_X \left[E(Y^0 \mid X, D=0) \cdot \frac{e(X)}{\pi} \right]$$

Bayes: $f(x \mid D=1) = e(x)f(x)/\pi$

$$= E_X \left[\frac{E[(1-D)Y \mid X]}{1 - e(X)} \cdot \frac{e(X)}{\pi} \right]$$

$$E[(1-D)Y \mid X] = (1 - e(X))E[Y^0 \mid X, D=0]$$

$$= E \left[\frac{(1-D)Y e(X)}{(1 - e(X)) \pi} \right]$$

law of total expectation

Line 3: it converts an average over the *treated* covariate distribution into one over the *whole* population.

Ten banks, split by size

Lecture 1's capital injections, now with a covariate: bank **size**.

Bank	Size	D	Y^0	Y^1	τ	observed Y
1	Small	1	10	12	2	12
2	Small	0	8	10	2	8
3	Small	0	9	11	2	9
4	Small	0	11	13	2	11
5	Small	0	12	14	2	12
6	Large	1	18	24	6	24
7	Large	1	19	25	6	25
8	Large	1	21	27	6	27
9	Large	1	22	28	6	28
10	Large	0	20	26	6	20

- Small banks are **rarely** rescued, 1 of 5; large banks **usually** are, 4 of 5. So $e(\text{Small}) = 0.2$ and $e(\text{Large}) = 0.8$.
- Small banks gain 2, large banks gain 6. Truth: $ATE = 4$, $ATT = 5.2$.
- Only the **bold** column is observed. Comparing rescued with not gives **11.2** — nearly three times the ATE .

The ATE rebuilds the population

Cell	N	e	injected	$\times 1/e$	controls	$\times 1/(1 - e)$
Small	5	0.2	1	$\times 5 = \mathbf{5}$	4	$\times 1.25 = \mathbf{5}$
Large	5	0.8	4	$\times 1.25 = \mathbf{5}$	1	$\times 5 = \mathbf{5}$
Total	10		5	10	5	10

- Bank 1 is the **only** small bank rescued, so it must stand in for all five. Its weight is $1/0.2 = 5$.
- Every weighted entry comes back to 5: each arm is blown back up to the **full population of ten**. That is the *pseudo-population*, and in it $D \perp\!\!\!\perp X$ by construction.
- Before weighting, small banks are 20% of the injected sample and 80% of the controls. After weighting, 50% of both — the imbalance that caused the bias is gone.

The ATT reweights controls only

The *ATT* wants the **treated** distribution. The four large and one small rescued banks are already a sample from it, so they keep weight 1. Only the controls move.

Cell	controls	$\times e/(1 - e)$	=	injected
Small	4	0.25	1	1
Large	1	4.00	4	4

- Multiplying the controls by $e/(1 - e)$ makes their count match the injected count **exactly** — last column.
- Bank 10 is the lone large control, and it must cover **four** rescued banks, so it carries weight 4. Push e toward 1 and there is nobody left to carry it: **this is where the ATT fails**, and why it needs only $e(X) < 1$ while the *ATE* needs $0 < e(X) < 1$.

Does it work?

$$\widehat{E}[Y^1] = \frac{1}{10} \left[\frac{12}{0.2} + \frac{24+25+27+28}{0.8} \right] = \frac{60+130}{10} = 19$$

$$\widehat{E}[Y^0] = \frac{1}{10} \left[\frac{8+9+11+12}{0.8} + \frac{20}{0.2} \right] = \frac{50+100}{10} = 15$$

$$\widehat{ATT} = \underbrace{\frac{12+24+25+27+28}{5}}_{\text{injected mean} = 23.2} - \underbrace{\frac{(8+9+11+12)(0.25)+20(4)}{4(0.25)+1(4)}}_{\text{controls} \times \text{odds} = 18.0}$$

- Truth: $ATE = 4$, $ATT = 5.2$. The naive SDO is $23.2 - 12 = \mathbf{11.2}$.
- IPW returns $19 - 15 = \mathbf{4}$ and $23.2 - 18.0 = \mathbf{5.2}$. Both exactly right.
- $ATT > ATE$ because large banks gain more, and large banks are where the injections went. **Selection on gains — the two estimands correctly disagree.**

Two ways to divide

Both estimators use the same weights $1/\hat{e}_i$. They differ **only in the denominator**.

$$\underbrace{\widehat{E}[Y^1]_{\text{HT}} = \frac{1}{N} \sum_i \frac{D_i Y_i}{\hat{e}_i}}_{\text{divide by } N}$$

$$\underbrace{\widehat{E}[Y^1]_{\text{Hájek}} = \frac{1}{\sum_i D_i / \hat{e}_i} \sum_i \frac{D_i Y_i}{\hat{e}_i}}_{\text{divide by the weights you got}}$$

- In the **population** they agree: $E[D/e(X)] = 1$, so the weights sum to N .
- In a **sample** they do not. $\sum_i D_i / \hat{e}_i$ is only an *estimate* of N , and it misses.
- Hájek's weights sum to one by construction, so it is always a genuine **weighted average of outcomes you actually observed**. HT carries no such guarantee.
- The default in Stata's `teffects ipw` and R's `WeightIt`.

Example 1

Four rescued banks. **Every one has lending growth** $Y = 10$, so the answer must be 10 for $\widehat{E[Y^1]}$. Here $N = 8$.

Bank	$\hat{e}$	weight $1/\hat{e}$	Y
A	0.50	2	10
B	0.50	2	10
C	0.50	2	10
D	0.25	4	10
sum		10	

$$\text{HT} = \frac{20 + 20 + 20 + 40}{8} = \frac{100}{8} = \mathbf{12.5}, \quad \text{Hájek} = \frac{100}{10} = \mathbf{10}.$$

- The weights sum to 10, not 8. HT divides by 8 anyway, returning $10 \times \frac{10}{8}$ — a number that appears nowhere in the data.

Example 2

Bank	$\hat{e}$	weight	share of weight	Y
A	0.50	2	3.6%	10
B	0.50	2	3.6%	10
C	0.50	2	3.6%	10
D	0.02	50	89.3%	24
sum		56	100%	

$$\text{HT} = \frac{1260}{8} = \mathbf{157.5}, \quad \text{Hájek} = \frac{1260}{56} = \mathbf{22.5}.$$

- D carries **89%** of the weight.
- **Normalization is necessary, not sufficient.** Check overlap separately: trim, or retreat to the *ATT* when the trouble is at $\hat{e} \approx 0$.

Propensity score matching

- Match each treated unit to controls with a **similar score**, then difference.
- **Caliper** matching: refuse any match farther than some tolerance (say 0.001 on the score), *discarding* the treated unit if none qualifies. **Radius** matching: average all controls inside the tolerance.
- Two problems with both:
 - ▶ The tolerance is **chosen by the researcher** (Huntington-Klein et al. 2021).
 - ▶ Dropping treated units **changes the estimand** — the *ATT* summarised the *original* treated sample.
- King & Nielsen, 2019: Two units can share an identical score $e(X_i) = e(X_j)$ despite having completely opposite covariate profiles $X_i \neq X_j$. Matching purely on the scalar score can actually increase local covariate imbalance compared to multivariate distance matching (e.g., Coarsened Exact Matching).

What the score does and does not buy

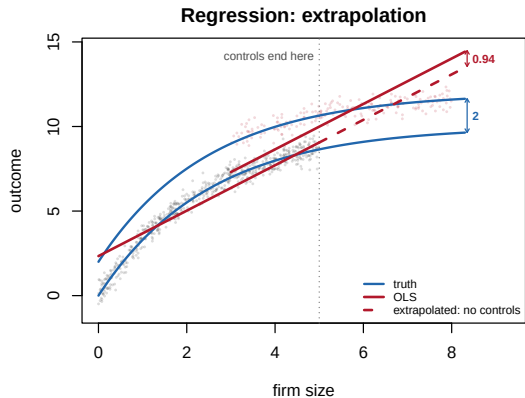
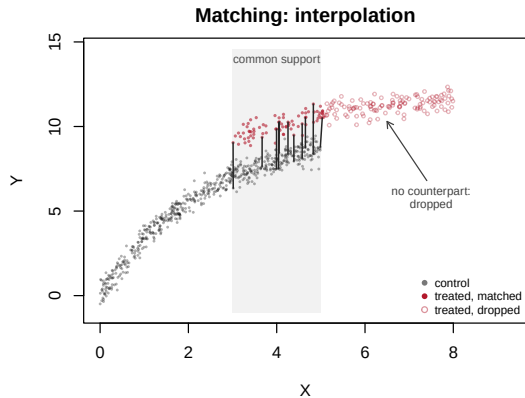
- It buys **dimension reduction**, not identification. If X omits a confounder, $e(X)$ omits it too.
- Standard regression silently hides lack of common support through linear extrapolation, whereas plotting $\hat{e}(X)$ immediately exposes severe positivity violations.
- A high pseudo- R^2 in a treatment model means covariates perfectly predict treatment assignment ($e(X) \rightarrow 0$ or 1). This breaks the positivity assumption and causes extreme IPW variance. Conversely, a low pseudo- R^2 indicates that treatment assignment is nearly independent of X (resembling a randomized control trial).
- IPW shifts the modeling assumption from $E[Y \mid D, X]$ (outcome equation) to $P(D = 1 \mid X)$ (treatment assignment equation).

2. Regression

Crossing to the other side of the map

- Everything so far has been an **interpolation** method: subclassification, matching, the propensity score. Each one refuses to compare a treated unit that has no counterpart. Common support is the binding constraint.
- Regression sits on the other side. It does not need common support at all, and so **the curse of dimensionality simply does not arise**.
- It buys that with a **functional form**. Where the data are silent, the fitted model speaks anyway: OLS imputes the missing counterfactual by **extrapolating** the outcome model into regions with no comparable units.
- That is the trade. Matching needs overlap; regression needs a correctly specified outcome model.
- Both still need unconfoundedness. Nothing in this section relaxes it.

Matching vs. regression



The bivariate case is easy; the multivariate case looks hard

- With one regressor,

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad \hat{\beta}_1 = \frac{\text{Cov}(Y_i, X_i)}{\text{Var}(X_i)}.$$

- The coefficient is the slope of the best-fit line, a **covariance scaled by a variance**.
- Now add covariates:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + \varepsilon_i.$$

- In a multivariate regression the coefficients are **not** simple scaled covariances. But the Frisch–Waugh–Lovell theorem (Frisch and Waugh 1933; Lovell 1963) says they are scaled covariances of something else.

Frisch–Waugh–Lovell in three steps

Step 1. Auxiliary regression:

$$X_{1i} = \gamma_0 + \gamma_2 X_{2i} + \cdots + \gamma_K X_{Ki} + f_i.$$

$\hat{X}_{1i}$ is the part of X_{1i} the other covariates explain.

Step 2. Residualization:

$$\tilde{X}_{1i} = X_{1i} - \hat{X}_{1i}.$$

The variation in X_{1i} that is *uncorrelated* with everything else — **partialling out**.

Step 3. Regress the outcome on the residual.

$$Y_i = \alpha + \beta_1 \tilde{X}_{1i} + \varepsilon_i, \quad \hat{\beta}_1 = \frac{\text{Cov}(Y_i, \tilde{X}_{1i})}{\text{Var}(\tilde{X}_{1i})}.$$

- $\hat{\beta}_1$ is **identical** to the coefficient on X_{1i} in the full regression.

FWL is an identity, not an approximation

Simulation: 4,000 workers, earnings on a treatment indicator d , age, and high-school GPA, with a true effect of \$1,500.

	(1) Multivariate OLS	(2) FWL
d	1503.201 (8.249)	1503.201 (8.287)
age	1.173 (2.023)	—
gpa	9.081 (4.615)	—
N	4000	4000

- The estimate sits 0.39 above the true 1500 — sampling noise.
- The standard errors do *not* agree, and that is not sampling noise.

Residualize both sides

- Another version residualizes the outcome too:

$$\tilde{Y}_i = Y_i - \hat{Y}_i \quad \text{from} \quad Y_i = \lambda_0 + \lambda_2 X_{2i} + \cdots + \lambda_K X_{Ki} + v_i, \quad (1)$$

$$\text{then} \quad \tilde{Y}_i = \beta_1 \tilde{X}_{1i} + \varepsilon_i. \quad (2)$$

- Both versions return **exactly the same** $\hat{\beta}_1$.
- But the double-residual version buys one more thing: **its residuals are the residuals of the full regression.**

Three regressions, one coefficient

	$\hat{\beta}_d$	s.e.
<i>original</i> : the controls explain 0.1% of earnings		
full OLS	1503.201	8.249
single, $Y \sim \tilde{D}$	1503.201	8.287
double, $\tilde{Y} \sim \tilde{D}$	1503.201	8.247
<i>expanded</i> : the controls explain 93%		
full OLS	1503.201	8.249
single, $Y \sim \tilde{D}$	1503.201	91.493
double, $\tilde{Y} \sim \tilde{D}$	1503.201	8.247

- Residualize only D : the residuals still carry everything age and GPA explain.

Why FWL is worth knowing

- **It tells you what variation identifies your coefficient.** $\hat{\beta}_1$ uses only the part of X_1 that the controls cannot explain.
- **It makes the estimate visualizable.** Residualize *both* sides against the controls and scatter $\tilde{Y}$ against $\tilde{X}_1$: a multivariate coefficient becomes a two-dimensional picture whose slope is $\hat{\beta}_1$.¹
- **It is the engine behind a great deal else.** Fixed effects, double machine learning, and the weighting results later in this section are all FWL in disguise.
- Angrist and Pischke (2009) call this **regression anatomy**, and it is the single most useful algebraic fact about OLS.

¹Filoso (2013) gives the worked proof and a Stata implementation.

Exogeneity, and the canonical OLS model

- Write the **canonical OLS model**:

$$Y_i = \alpha + \delta D_i + \gamma X_i + \varepsilon_i.$$

- For $\hat{\delta}$ to be causal, D_i must be *exogenous* $\Rightarrow$ Imbens and Rubin (2015) put it plainly: exogeneity “combines unconfoundedness with functional form and constant treatment effect assumptions that are quite strong, and arguably unnecessary.”
- Three assumptions, not one. Unconfoundedness you have already met and argued for. The other two arrived silently, in the act of writing down the equation.

Taking conditional expectations

- Take expectations of the canonical model separately for the treated and the control group:

$$E[Y_i \mid D_i = 1, X_i] = \alpha + \delta E[D_i \mid D_i = 1, X_i] + \gamma_1 E[X_i \mid D_i = 1, X_i]$$

$$E[Y_i \mid D_i = 0, X_i] = \alpha + \delta E[D_i \mid D_i = 0, X_i] + \gamma_1 E[X_i \mid D_i = 0, X_i]$$

- Now replace Y with Y^1 or Y^0 as appropriate — and do the same for X .
- Why would X have potential outcomes? **If $X^1 \neq X^0$, then D causes X , and X is an outcome, not a covariate.** That is the bad-control problem.
- And allow the covariate to matter differently in the two worlds, γ_{11} against γ_{01} :

$$E[Y_i^1 \mid D_i = 1, X_i] = \alpha + \delta E[D_i \mid D_i = 1, X_i^1] + \gamma_{11} E[X_i^1 \mid D_i = 1, X_i^1]$$

$$E[Y_i^0 \mid D_i = 0, X_i] = \alpha + \delta E[D_i \mid D_i = 0, X_i^0] + \gamma_{01} E[X_i^0 \mid D_i = 0, X_i^0]$$

What $\gamma_{11} \neq \gamma_{01}$ actually means

- Let Y^1 be earnings with a college degree, Y^0 earnings without one, and X be age.
- Then γ_{11} is the age–earnings profile for **college graduates** and γ_{01} is the age–earnings profile for the same people **had they not gone**.
- $\gamma_{11} \neq \gamma_{01}$ **is** heterogeneous treatment effects — the return to treatment varying with X .
- In finance the same statement is unremarkable: the effect of an index inclusion, a rating change, or a governance shock varies with firm size. If it does, $\gamma_{11} \neq \gamma_{01}$.

Heterogeneous treatment effect bias

- OLS on the canonical model is a difference in conditional means:

$$\hat{\delta}_{OLS} = E[Y_i^1 \mid D_i = 1, X_i^1] - E[Y_i^0 \mid D_i = 0, X_i^0].$$

- Substitute the two expressions from the previous slide.

$$\hat{\delta}_{OLS} = \delta + \underbrace{\left(\gamma_{11}E[X_i^1 \mid D_i = 1, X_i^1] - \gamma_{01}E[X_i^0 \mid D_i = 0, X_i^0] \right)}_{\text{heterogeneous treatment effect bias}}$$

- This is what the canonical OLS model identifies. Not δ . δ plus a term that has nothing to do with omitted variables.
- Notice what is *not* in the bias term: any unobserved confounder. We assumed unconfoundedness holds perfectly and there is nothing left to omit.

The two assumptions you signed up for

The bias term vanishes if and only if **both** of the following hold.

1. **Covariates are not outcomes.** $X^1 = X^0 = X$: the covariates do not respond to treatment.
 2. **Treatment effects are homogeneous in X .** $\gamma_{11} = \gamma_{01} = \gamma_1$: the return to treatment does not vary with the covariates.
- Under A1 and A2, $\hat{\delta}_{OLS} = \delta$. Fail either one and the canonical model is biased.
 - **A1 has no repair.** Do not control for a collider or a mediator $\Rightarrow$ Lecture 2!
 - **A2 does have a repair**, and it is a change to the specification, not to the data. That is the next slide.
 - Note how strong A2 is. Neither matching nor the propensity score ever required it: they average heterogeneous effects, they do not assume them away.

Regression adjustment: Oaxaca's answer

- The problem was never OLS. The problem was that **one specification** — additive and separable.
- The **Kitagawa–Oaxaca–Blinder decomposition** splits a mean gap into a part attributable to differences in *endowments* (education, experience) and a part attributable to differences in the *returns* to those endowments.
- Słoczyński (2015, 2020) showed the decomposition is also a **treatment effects estimator** — and that it identifies both the *ATE* and the *ATT*. This is what “regression adjustment” means.

Deriving the specification

- Write both potential outcomes, letting X act differently on each:

$$Y_i^0 = \alpha + \gamma_{01}X_i + \varepsilon_i$$

$$Y_i^1 = \alpha + \gamma_{01}X_i + \delta D_i + \gamma_{11}X_i \cdot D_i + \varepsilon_i$$

- Take conditional expectations, then form $\delta = E[Y_i^1 \mid D_i, X_i] - E[Y_i^0 \mid D_i, X_i]$:
$$\begin{aligned} &= (\alpha + \gamma_{01}E[X_i \mid D_i=0, X_i] + \delta + \gamma_{11}E[X_i \cdot D_i \mid D_i=1, X_i]) - (\alpha + \gamma_{01}E[X_i \mid D_i=0, X_i]) \\ &= \delta + \gamma_{11}E[X_i \cdot D_i \mid D_i = 1] \end{aligned}$$
- The γ_{01} terms cancel because the *same* coefficient appears in both potential outcomes. The interaction is what remains — and it is exactly the term the canonical model omitted.

The regression adjustment estimator

To identify the *ATE* and the *ATT* under unconfoundedness **with heterogeneous effects**, run

$$Y_i = \alpha + \delta D_i + \gamma_{01} X_i + \gamma_{11} D_i \cdot X_i + \varepsilon_i.$$

- Interact the treatment with **every** covariate. Then

$$\widehat{ATE} = \hat{\delta}, \quad \widehat{ATT} = \hat{\delta} + \hat{\gamma}_{11} E[X_i \mid D_i = 1],$$

where $E[X_i \mid D_i = 1]$ is the covariate mean **in the treated group only**.²

- The difference between the two estimands is now visible and mechanical: the same coefficients, evaluated at a different covariate distribution.

²The *ATE* result requires X to be demeaned in the full sample. If you interact with terms that are not mean-zero — squares, cross-products — add them back: $\widehat{ATE} = \hat{\delta} + \sum_k \hat{\gamma}_{1k} \bar{X}_k$.

Does it work?

Simulation: 5,000 workers, 5,000 with a college degree; two exogenous covariates, age and GPA, with **different means in the treated and control groups**; both recentred, plus their squares and their interaction. Then

$$Y^0 = 15000 + 10.25 \text{ age} - 10.5 \text{ age}^2 + 1000 \text{ gpa} - 10.5 \text{ gpa}^2 + 500 (\text{age} \cdot \text{gpa}) + e$$

$$Y^1 = Y^0 + 2500 + 100 \text{ age} + 1100 \text{ gpa}$$

- Because age and GPA are recentred, $E[Y^1 - Y^0] = \$2,500$ exactly. The *ATT* is **not** \$2,500: the treated group has a different age and GPA composition, and the effect varies with both.
- The canonical model here controls for age, age², GPA, GPA² and their interaction. Every covariate needed for unconfoundedness is in it. Nothing is omitted.

When is the canonical model actually wrong?

Simulation, varying the **share treated**:

Share treated	0.20	0.35	0.50	0.65	0.80
True <i>ATE</i>	2500	2500	2500	2500	2500
True <i>ATT</i>	1813	1942	2070	2199	2328
Canonical OLS	2097	2310	2500	2689	2902
Regression adj. (<i>ATE</i>)	2500	2500	2499	2499	2499
Regression adj. (<i>ATT</i>)	1812	1941	2069	2198	2327

- Regression adjustment recovers **both** estimands at every split.
- At exactly 50/50 it lands on the *ATE* — **by coincidence of the design, not by validity.**

What regression adjustment does not buy

- Regression adjustment removes the **constant-effects** assumption. It does not remove the **functional form** assumption, and cannot.
- Inside the model is a model of the missing potential outcome. In the simulation the true Y^0 contained age, age², GPA, GPA² and age · GPA — so the treatment had to be interacted with **all five**.
- Drop just one interaction and it breaks. With the treatment effect also varying in age · GPA, the fully interacted model returns 2576 against a true *ATE* of 2575; omitting the single $D \cdot (\text{age} \cdot \text{gpa})$ term returns 2368.
- **The functional-form requirement replaces the common-support requirement. It is not free.** You have swapped a checkable condition for an uncheckable one.
- This is the honest summary of the trade made at the start of the section.

Three things wrong with the canonical model

1. **It assumes constant treatment effects.** Neither matching nor the propensity score ever asked for this. Regression adjustment removes it, at the cost of interacting the treatment with everything.
2. **Its weights are wrong.** Słoczyński (2022) shows the coefficient is $w_1 \cdot ATT + w_0 \cdot ATU$, with each group's weight **inversely** proportional to its size; **it is generally none of the three.**
3. **It can only ever deliver one number.** There is one treatment indicator, so there is one coefficient. Regression adjustment delivers the *ATE* and the *ATT* from the same fit.

3. What Matching Cannot Do

Matching creates no new variation

- Matching **does not introduce exogenous variation**. It reorganizes and reweights the data you already have, using observables you already measured.
- Consequence: if your OLS regression was biased by endogeneity, **the matched estimate is biased too**.
- Contrast this with the designs still to come:
 - ▶ **IV** imports variation from outside the model.
 - ▶ **RDD** exploits assignment that is locally as good as random.
 - ▶ **DiD** differences out fixed unobservables.
- Matching belongs to a different category. It is a *conditioning strategy*, like regression — not a research design.

What it cannot fix

Matching is powerless against:

- **Unobserved confounders.** You cannot match on what you cannot measure.
- **Reverse causality.** Matching on X says nothing about whether Y caused D .
- **Measurement error** in treatment or outcome.
- **Bad controls.** Matching on a post-treatment variable is exactly as damaging as controlling for it — Lecture 2 applies unchanged, and the collider F of Lecture 3a, §2, Confounders, Covariates, and Colliders, is the warning.

You cannot match on what you cannot measure.

So what is it good for?

Given all that, matching still earns its place:

1. **A non-parametric benchmark.** It estimates the effect without assuming a functional form. If matching and OLS disagree sharply, your specification is doing more work than you thought.
 2. **Common-support discipline.** Matching refuses to compare units with no counterpart. Regression silently extrapolates into regions with no data.
 3. **Diagnostic value.** The propensity score distribution reveals overlap problems that a regression table hides completely.
 4. **Robustness when the functional form is wrong.** Matching can outperform a misspecified regression in finite samples.
- Use it as a *complement* to regression and a check on it — not as an identification strategy.

4. The NSW Demonstration

The 1970s empirical crisis

- The methods in this lecture were not adopted because they were elegant. They were adopted after a crisis of confidence in observational work.
- Job-training programs in the 1970s were evaluated observationally, and the estimates were all over the place — depending, apparently, on the analyst.
- The **National Supported Work (NSW) Demonstration** was different: it *randomized* participation. So the experimental effect was known.
- That created something rare and valuable: a **benchmark**. You could now ask whether observational methods, applied to the same treated units, recover the experimental answer.
- Everything in this lecture gets graded against it.

What LaLonde did

- LaLonde (1986) ran the decisive test. He took the NSW experiment, **threw away the experimental control group**, and replaced it with comparison groups built from survey data (CPS and PSID).
- He then applied the standard observational toolkit of the day and compared the results to the experimental benchmark of roughly **\$1,800** in annual earnings gains.
- The observational estimates ranged from a fraction of the benchmark to **the wrong sign** — and which one you got depended heavily on how the comparison group was built.
- The paper was devastating and hugely influential. It is a major reason the profession moved toward IV, RDD, and DiD — the other river.

Dehejia and Wahba: not so fast

- **Dehejia and Wahba (1999, 2002)** revisited LaLonde's data with propensity score methods.
- Their finding: the observational estimates *did* recover the experimental benchmark reasonably well — **once common support was imposed**.
- The key diagnosis: much of LaLonde's failure was an **overlap** problem, not only a confounding problem. The CPS and PSID comparison groups contained enormous numbers of people utterly unlike the NSW participants.
- Discard the non-comparable units and the methods perform far better.
- This is why the distinction of Lecture 3a, §3, Identification Assumptions, matters so much. Two different assumptions were failing, and only one of them is checkable.

Smith and Todd: not so fast either

- **Smith and Todd (2005)** re-examined the Dehejia–Wahba result and found it fragile.
- The success was **sensitive to the subsample** analysed and to the **propensity score specification** chosen. Reasonable alternative choices undid it.
- Their charge is uncomfortable and worth stating plainly: the specification that worked may have been found *because* the experimental answer was already known.
- You cannot run that search in real research. In a genuine application there is no benchmark to check against — that is the entire point of doing the study.

The honest summary

- Selection on observables **can** work — when overlap is genuine and the confounders are truly measured.
- Neither condition is verifiable in the setting where you need them:
 - ▶ Unconfoundedness is **untestable**, always.
 - ▶ Common support is **checkable**.
- So the practical lesson is asymmetric: do the check you can do, and argue explicitly for the assumption you cannot.
- **A useful exercise:** the NSW data are freely available. Apply every method in this lecture and compare them to the \$1,800 benchmark. Cunningham's chapter walks through it, and it is the fastest way to develop instincts about when these methods hold up.

Wrapping Up

Summary [Part 1]: the score and what it does

- **The propensity score** reduces X to one number and defeats the curse of dimensionality for *conditioning*. It does nothing for **identification** — unconfoundedness is still assumed, never delivered.
- **Judge the score by balance achieved, not by fit.** A high pseudo- R^2 with bad balance is a bad score; the reverse is fine. The whole stage is blind to Y , which is what makes the specification search legitimate.
- **Balance means the distribution, not the mean.** Standardized differences *and* variance ratios. Our Volatility covariate had $SMD = -0.010$ and a variance ratio of 2.27.
- **IPW** reweights the sample into a pseudo-experiment: $1/\hat{e}$ for treated and $1/(1 - \hat{e})$ for controls builds the population; $\hat{e}/(1 - \hat{e})$ on controls alone rebuilds the *treated* distribution. Watch the tails, and normalize.
- **Propensity score matching** is the one use with a serious case against it: King and Nielsen (2019) show it targets complete randomization when you could have had blocking, so pruning can *increase* imbalance.

Summary [Part 2]: the verdict

- **Regression is a conditioning strategy too**, not a rival family. FWL shows it uses only the **residual variation in D** — and Słoczyński's weights show that the coefficient this produces is generally neither the *ATE* nor the *ATT*.
- **Matching creates no new exogenous variation**. If OLS was endogenous, matching is endogenous. It cannot fix an unobserved confounder, reverse causality, or a bad control.
- **NSW**: these methods can work when overlap is real and the confounders are measured — and you can only ever verify one of those two.
- Next lecture: designs that *create* variation instead of assuming it away.

A checklist for your own paper

1. State the **estimand** before the estimator. *ATE*, *ATT*, or *ATU* — and say why the policy question needs that one.
2. Defend the **assignment mechanism** in institutional terms. Name the confounders and say how you measured them.
3. Justify every covariate as **pre-treatment**. Any post-treatment control needs an argument, not a footnote.
4. **Plot the overlap**. Report what you trimmed and how the estimand changed.
5. Report **balance**, not fit: standardized differences and variance ratios, before and after.
6. Report **matching and regression**. If they disagree, diagnose whether it is weights or extrapolation.
7. Run a **sensitivity analysis**, and calibrate the threshold against an observed covariate.
8. Run a **placebo** on a pre-treatment period.
9. Say plainly what would break the design — and what unobservable would have to exist.

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